Exam I Information (Oct 28, 9.00-12.00)

Multiple-choice questions

Lectures covered - Introduction, Algorithm efficiency analysis, Linear-time sorting and Order statistics, Graph traversals and connectivity, Network flows and applications, Matching in Bipartite graphs.

Summation formulas are given.

"ANY FORM OF EXAMINATION MISCONDUCT IS CONSIDERED A SERIOUS DISCIPLINARY OFFENCE."

Instructions

1. This exam consists of 46 questions (61 points) – 11 pages (cover included)
2. Mark X in the provided answer sheet with a blue-ink or black-ink pen.
3. Write your name and student ID in this page and the answer sheet page.
4. Do not separate the answer sheets from the exam paper.
5. If you believe that some exam question is unclear or there are any doubts, write notes explaining your reasoning to solve the problem in the answer sheet.

Permissions

1. A calculator is not allowed.
2. A student can bring in a sheet of A4 paper note and submit it with the exam paper.
3. Any other documents and textbooks are not allowed.

Study Guide (1)

Terminology, definitions, principles, and concepts (34 points)

- Algorithm designs, algorithm efficiency analysis, and asymptotic notations.
- Analyzing $C(n)$ of given algorithms
- SCC, Residual graph, Max-flow-min-cut theorem
- Max-flow solution to maximum matching
- Blocking pair in marriage matching
- Properties of M-augmenting path
- Kuhn-Munkres theorem
- Label improvement in Hungarian algorithm

Study Guide (2)

Work out algorithms to solve problems

- Lomuto partitioning (3 pts)
- DFS and Topological sorting (3 pts)
- Strongly connected component (3 pts)
- Ford-fulkerson method (6 pts)
- Feasible circulation (2 pts)
- Maximum weight matching (2 pts)
- Stable marriage (4 pts)
- Survey design network formulation (4 pts)